AP PRECALCULUS

Unit 1 • Worksheet 1.2A • Answer key

Worksheet 1.2A • Rates of Change

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the average rate of change of $f(x) = 2x^2 - 3$ on $[1, 4]$?

(A) 10

(B) 15

(C) 30

(D) 8

Solution

Answer: (A). $f(4) = 29$, $f(1) = -1$: $\frac{30}{3} = 10$.

Question 2

NO CALCULATOR

What is the average rate of change of $g(x) = x^3 - 2x$ on $[-1, 2]$?

(A) 3

(B) $\frac{5}{3}$

(C) 1

(D) -1

Solution

Answer: (C). $g(2) = 4$, $g(-1) = 1$: $\frac{3}{3} = 1$.

Question 3

NO CALCULATOR

What is the average rate of change of $h(x) = \dfrac{12}{x}$ on $[2, 6]$?

(A) -1

(B) -4

(C) 1

(D) $-\frac{1}{2}$

Solution

Answer: (A). $h(6) = 2$, $h(2) = 6$: $\frac{2-6}{4} = -1$.

Question 4

NO CALCULATOR

The average rate of change of $f(x) = x^2 + kx$ on $[0, 2]$ is 5. What is k?

(A) 1
(B) 5
(C) 3
(D) 7

Solution
Answer: (C). $\frac{f(2)-f(0)}{2} = \frac{4+2k}{2} = 2 + k = 5$, so $k = 3$.

Question 5 (Free response)

NO CALCULATOR

Let $f(x) = x^2 - 6x + 10$.

(a) Find the average rate of change of f on $[0, 3]$ and on $[3, 6]$.
(b) Explain what the signs of your answers tell you about f on each interval.
(c) Find a value $b > 0$ such that the average rate of change of f on $[0, b]$ is 0.

Solution
(a) $f(0) = 10$, $f(3) = 1$, $f(6) = 10$. On $[0, 3]$: $\frac{1-10}{3} = -3$. On $[3, 6]$: $\frac{10-1}{3} = 3$.
(b) Negative: on average f decreases on $[0, 3]$. Positive: on average f increases on $[3, 6]$.
(c) We need $f(b) = f(0) = 10$: $b^2 - 6b = 0$, so $b = 6$.